"Formal Frameworks for Semantics and Pragmatics"

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1. Frameworks as theoretical vocabularies

Like many sciences, semantics and pragmatics are marked by a great abundance of *theoretical vocabulary* (jargon), introduced without explicit definitions. Different "theories" or "approaches" often use different theoretical vocabulary—differences like that are what we have in mind when we talk of different 'frameworks'.

Often, different frameworks use the same theoretical vocabulary with different meanings, or at least meanings that aren't *obviously* the same. Some culprits:

- 'Semantic value', 'denote', 'stand for', 'refer to', 'express'…
- 'Truth conditions'
- 'Context'
- 'Presuppose'
- 'Sentence' and 'expression'

This means that apparent disagreements are not always real. We'll need to be cautious!

2. Connections from T-terms to O-terms

The theories/approaches include "bridge laws" which link up the theoretical vocabulary ('T-terms') with other, non-proprietary vocabulary ('O-terms'). The bridge laws are often less explicit and rigorously worked out than the "pure" part of the theory. Still, presumably the point of the whole enterprise is to answer questions that posed using O-terms.

Let's distinguish three groups of possible bridge laws.

Group A: "Metasemantic" bridge laws. The sorts of principles that someone might reasonably regard as central for answering metaphysical questions expressed using the T-terms, like 'What is it for a sentence to express a proposition?' or 'In virtue of what do names denote whatever they denote?'

1. Connections to ordinary speech-act verbs like 'say', 'assert', 'imply', 'suggest', 'ask', 'command', 'propose', 'represent' (as in 'he represented himself as an expert'), 'commit' (as in 'she committed herself to quitting smoking'). Also verbs like 'refer', 'call', 'name' (applied to speakers or utterances rather than linguistic expressions).
    - Examples: 'When a sentence <u>expresses</u> the proposition that P in a community, speakers who utter it sentence will generally say [/assert/ be believed to have said / commit themselves to maintaining / propose that their interlocutors should accept] that P'.

2. Connections to ordinary *propositional-attitude* verbs like 'believe', 'expect', 'want', 'know', 'prefer', 'be confident'.

    - Example: 'When a sentence <u>expresses</u> the proposition that P in a community, most members of the community know that typically, people who utter the sentence believe that P'.

3. Connections to the vocabulary of "folk semantics"—words like 'mean', 'refer to', and 'true', as applied to linguistic expressions (as opposed to speakers or utterance events).

    - Example: 'When a sentence's semantic value is the set of worlds at which P, it means that P'.
    - In my view it's a bad idea to take much for granted here. The flowering of formal semantics since the 60s has been partly enabled by people *not* insisting that theories connect up straightforwardly to ordinary talk about "what things mean".

4. Connections (actual and hoped-for) to the proprietary theoretical terminology of *other* sciences, especially cognitive psychology.

    - Example: 'When a sentence expresses the proposition that P, the first step in a hearer's processing of an utterance of that sentence is to construct an internal mental representation with the content that P'.

Group B: "Operational" bridge laws. In practice, you can do lots of great semantics in a way that's neutral about metasemantics, relying instead on bridge laws using O-terms like 'anomalous', 'felicitous', 'interchangeable', etc., which classify people's (one's own or experimental subjects') reactions to examples sentences. The bridge laws that let us extract such predictions are typically far from explicit: there is a lot of unarticulated know-how.

- We can get surprisingly far with predictions about *similarity and difference*: you can just eyeball a theory and see that it's saying the same sort of thing about two sentences, so that without further supplementation it can't explain some difference in our reactions to the sentences.

Group C: "User-oriented" bridge laws. Even if they have no interest in language for its own sake, philosophers have good reason to care about semantics and pragmatics. Philosophers need to use natural languages to articulate their views to one another (and to themselves), to formulate arguments, to raise questions. Fine linguistic distinctions often matter in a way they rarely matter in everyday life, and there are lots of ways things can go wrong: e.g. we can commit fallacies of equivocation, or misapply argument-forms that are only of limited validity. Semantics and pragmatics can help here thanks to bridge-laws involving O-terms like 'valid', 'inconsistent', and 'ambiguous'. (Also simply by highlighting similarities and differences.)

3. 'Micro-theory' and 'macro-theory'

A common, fruitful structure is to have the overall theory break into two parts. Just for today we will use the not-perfectly-apt labels 'micro-theory' and 'macro-theory' for the parts, since existing labels like 'semantics' and 'pragmatics' have lots of other associations.

The hallmark of the *micro-theory* is a kind of parallel treatment for all expressions of the language. Often this takes the form of generalisations about how the "semantic values" of complex expressions relate to those of their constituents. From the standpoint of this theory, sentences are just another kind of complex expression; what's said about sentences will be broadly isomorphic to what's said about, e.g., noun phrases.

By contrast, the *macro-theory* will traditionally give a special role to sentences, or multi-sentence "discourses". (It'll take an interest in smaller units only insofar as there's a practice of sometimes uttering them in isolation.) In some cases the macro-theory will consist entirely of bridge laws connecting the theoretical terms of the micro-theory to O-terms, like those considered above. But in many cases the macro-theory will bring in new T-terms of its own:

> When someone utters a sentence whose <u>semantic value</u> is a certain set S of worlds and this utterance is <u>accepted</u>, the <u>scoreboard</u> of that conversation changes by intersection of its first co-ordinate with S.
>
> When someone utters a sentence whose semantic value is a partial function from worlds to truth values that is defined only on worlds at which P, the result will be a <u>crash</u> unless the <u>presupposition</u> that P can be <u>accommodated</u>'.

Often there's a debate whether some phenomenon (e.g. scalar implicatures) is best explained by complicating the micro-theory (and just minimally modifying the macro-theory to reflect the intended bridge laws for the new micro-T-terms), or entirely within the macro-theory.

4. Model theory in semantics

The role of model theory in most work in formal semantics makes the path from theoretical claims to O-terms even more indirect. Models are defined as mathematical objects of some sort, that give rise in some mathematically natural way to one or more "domains"; much theorising involves giving explicit mathematical definitions of relations between expressions and elements of these domains (and perhaps various other relata such as assignment functions).

Often the model-relativity of the definitions is not made explicit. But you can tell that it's implicitly there when you see semanticists saying things like 'Let D be the set of all individuals that exist in the real world' (Heim and Kratzer p. 15). Taken at face value, this conflicts with standard set theory, which entails that no set is a member of itself.

- One unattractive alternative to model-relativity is to say that semanticists are only concerned with a limited fragment of human linguistic ability that excludes our capacity to, e.g., say that *absolutely everything* depends on God by uttering the words 'Absolutely everything depends on God'.

Bridging the gap from model-relativised theoretical claims to non-model-relativized O-claims, e.g. about what you could say using a given sentence, is extra-hard!

Suggestion: theories expressed in model-theoretic terms function as technically convenient *surrogates* for theories formulated in richer metalanguages.

- *Example:* a theory about a model-relative 'semantic value' function that maps names to one set D_e, sentences to a different set D_t, noun phrases to $D_t^{D_e}$, etc., can be regarded as a surrogate for a theory stated in a *higher-order* metalanguage. The higher-order semantic theory will have an expression 'expresses$_t$' that can combine with a name (for an object-language sentence) and a sentence to make a sentence, and an expression 'expresses$_{e\to t}$' that can combine with a name (for an object language noun-phrase) and a predicate to make a sentence, etc.